CODE N' CREATE | LED CIRCUIT CHALLENGES

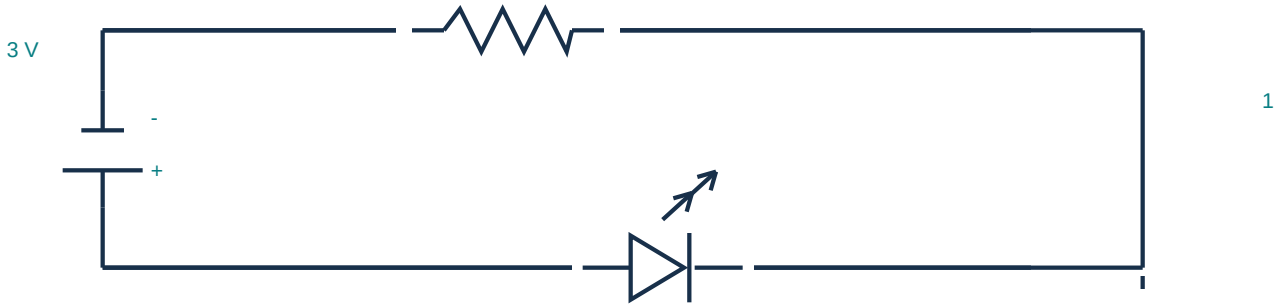
Tuesday Lab | Grades 7-8 | Name: _____

Equipment: 2 AA cells (3 V), 5 mm LEDs, 4-band resistors, multimeter, leads

SAFETY: Never power an LED without a current-limiting resistor.
Power OFF when changing wiring or measuring resistance. Teacher checks each build.

Symbols: zigzag = resistor; triangle-like diode with outward arrows = LED; long/short lines = battery.

Challenge 1 - One resistor, one LED



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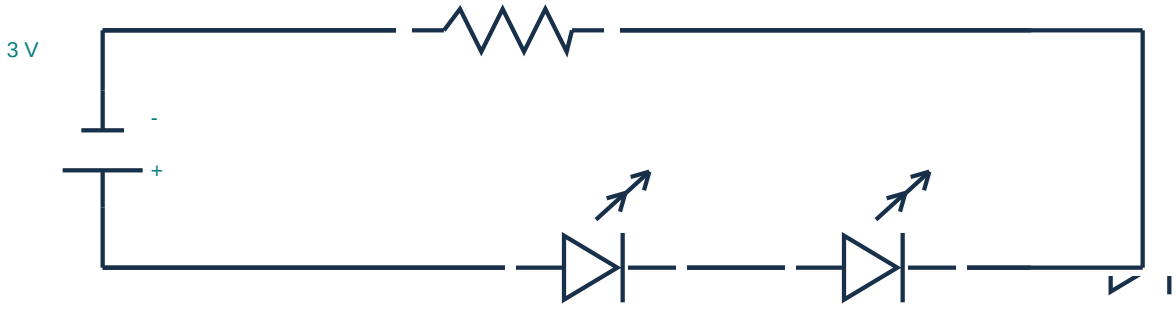
1. Record resistor color bands and calculated value: _____

2. Predict and record LED brightness: _____

3. What happens if you reverse the LED? Why? _____

Challenge 2 - One resistor, two LEDs in series



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4. Predict: Will both LEDs light with 3 V? Record what actually happens.

5. Compare brightness with Challenge 1. Explain your observation.

CODE N' CREATE | CIRCUIT INVESTIGATIONS

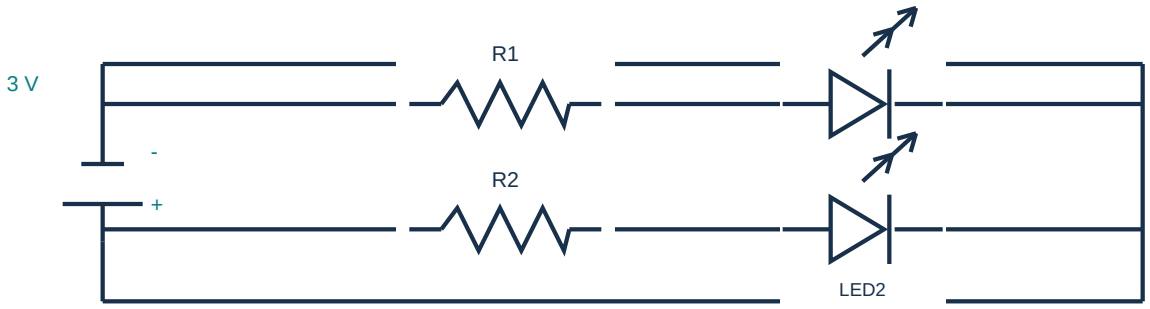
Grades 7-8 | Name: _____

Safety investigation - LED without a resistor (DO NOT BUILD)

Predict what could happen if a 3 V battery is connected directly to a 5 mm LED.

Explain how a resistor protects an LED: _____

Challenge 3 - Two parallel branches (R + LED in each)



6. Which branches light? Compare LED brightness and explain.

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7. Disconnect one branch with power OFF. Predict what happens to the other.

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8. Measure each resistor separately with power OFF: R1 ____ ohms; R2 ____ ohms.

Exit ticket: One thing I learned about current-limiting resistors: